Sahaj Bhadja

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Summary

Software Engineer with 2+ years shipping **Python** and **TypeScript** backends, multi-source integrations, and production **LLM** workflows, plus a deployed **RAG** app with incremental ingestion, building developer-facing tools with product judgment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **LLM orchestration** as fault-tolerant FastAPI background workers with explicit job-state transitions and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Built **integration pipelines** syncing FastAPI services, Supabase, PostgreSQL, SQL Server, and cloud storage with incremental loads and reconciliation, running 5K+ monthly jobs at 99%+ reliability for 500+ users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a provider-agnostic **LLM layer** across OpenAI, Anthropic, and Qwen with dynamic model switching, OCR fallback, and per-call token and cost tracking, powering prompt chains for 10K+ daily requests.
- Shipped a React and Flask grading product for 150+ instructors with secure PDF, DOCX, and ZIP **ingestion** and concurrent batches of 1K+ files, cutting feedback turnaround 60% at 90%+ consistency.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Normalized messy JSON, CSV, and Parquet data from multiple sources into unified formats with AWS Glue and S3, consolidating activity logs for analytics and **API consumption** and boosting pipeline efficiency 68%.
- Optimized large-scale **API integrations** with pagination, field filtering, and caching, improving response times 30% and cutting technical debt 20%, while collaborating with Microsoft and GitHub stakeholders in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight **TypeScript** and Node.js microservices behind REST APIs for authentication, resources, billing, leads, and email and SMS notifications, syncing 2M+ records across locations through real-time WebSocket updates.
- Automated Jest, Mocha, and Selenium tests in Jenkins **CI/CD** with Docker packaging, reaching 95%+ build validation, and used usage analytics to drive changes that raised lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Gathered client requirements, mapped 12+ **integration points** into an end-to-end architecture, and presented it before building Python Flask APIs for bookings, pricing, coupons, and payments with Firebase sync.
- Integrated third-party payment providers through idempotent **webhooks** for refunds, disputes, and chargebacks, cutting manual handling 70%, and built hourly Databricks ingestion over 50K+ booking and payment events.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, Webhooks, WebSockets, Microservices, Background Workers
AI & Retrieval: LLM Orchestration, RAG, Embeddings, FAISS, Pinecone, Prompt Chaining, Multi-Model Routing, Claude, GPT-4o
Data & Integrations: PostgreSQL, Supabase, MongoDB, Firebase, ETL Pipelines, Incremental Loads, Data Validation, AWS Glue
Cloud & DevEx: AWS, Azure, GCP, Docker, CI/CD, GitHub Actions, Vercel, Technical Documentation
Practices: Claude Code, Code Reviews, Agile

Projects

Workspace Memory SDK | *Python, TypeScript, FastAPI, PostgreSQL, pgvector, Claude*

- Designed Python and TypeScript **SDKs** over a memory API that syncs Slack, Gmail, and Notion incrementally, embeds chunks into pgvector with freshness-weighted hybrid retrieval, and tracks recall@k on an **eval** set.

Big-Brain Live | *LLaMA-2, Pinecone, FAISS, Clerk, Convex, PostgreSQL*

- Built a deployed **RAG** app with Clerk auth and an ingestion pipeline that incrementally indexes 50K+ documents into Pinecone and FAISS with metadata, cutting retrieval latency 40% at 95% query accuracy.

Chat2SQL Code | *TypeScript, Java, PostgreSQL, Docker*

- Extended an open-source **developer tool** in Java and TypeScript into an AI-assisted SQL workspace for 10+ databases, generating schema-aware queries at 82% accuracy with Docker-based local setup.